

Luke Serrano

 Luke Serrano |  lukeserrano0@gmail.com |  (618) 334-5046

EXPERIENCE

Sandia National Laboratory Internship

Jan 2024 - Present

Navigation, Guidance, and Control Engineer

Albuquerque, NM

- Analyzed mathematics in INS algorithm and Kalman based estimation filter to explain undesired navigator behavior from a given accelerometer and gyroscope dataset
- Worked within the Simulink simulation and MATLAB post processing repositories to improve navigator verification framework
- Modified autogenerated XML messaging framework, C++ embedded, and Java GUI code to add the ability to modify ACS parameter via FC interactive GUI. Tested functionality on HWIL setup
- Worked with GitLab submodules, rebasing, mixed resets, squashed commits, merge requests, and resolving merge conflicts to push development code up for CI and review
- Performed valve pressure timing analysis to better characterize physical hardware for simulation
- Proved ability to work within multi-language framework, dive into mathematical theory, work within a Git workflow, and contribute meaningfully to an NGC team within 7 months of employment

Missouri S&T Satellite Research Team (M-SAT)

Aug 2022 - Present

Missouri University of Science & Technology

Rolla, MO

- Chief Engineer of MR/MRS SAT mission
- Building interprocess communication software framework to integrate simulation suite and NGC algorithm to verify NGC requirements via simulation
- Manage NGC embedded and simulation repositories
- Communicate with 10 subsystem leads to ensure proper direction and progression of satellite development

Northrop Grumman Internship

June 2023 - Aug 2023

Northrop Grumman Space Sector

Gilbert, AZ

- Specialty Systems Engineer Intern at satellite manufacturing facility
- Exposure to industry with requirements, a paying customer, deadlines, and expected mission success

EDUCATION

Missouri University of Science & Technology

Fall 2022 - Spring 2025

B.S. in Aerospace Engineering

GPA: 3.8

Orbital Mechanics II; Aerodynamics II; Flight Dynamics and Controls I; Spacecraft Design Lab

Southwestern Illinois College

Jun 2021 - Aug 2022

Pursued Associates in Science

GPA: 3.8

SKILLS

- | | | |
|---------------------------------|-----------------------|--------------|
| - C++ | - MATLAB | - Java |
| - GitLab Development | - Quick Learner | - Leadership |
| - Circuitry Board Communication | - Strong Communicator | - Discipline |